

Predictive Monitoring of Business Processes: A Case Study Using a Cumulative Fact Table Model

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Abstract— Predictive Business Process Monitoring (BPPM) is an increasingly used technique to anticipate future events within organizations, based on the analysis of historical and real-time data. This article explores the application of a cumulative fact table model in a data warehouse to improve the accuracy and speed of predictions within the context of BPPM. Through a case study in the logistics sector, the proposed model significantly reduced the error rate in delivery time predictions and enabled faster anomaly detection compared to traditional methods. This approach has proven particularly effective for real-time monitoring of business processes, while also highlighting the need for a robust data infrastructure for its implementation.

Keywords— Predictive Monitoring, Business Processes, Cumulative Fact Table, Data Warehousing, Business Intelligence, Predictive Analytics

I. INTRODUCTION (HEADING 1)

In an increasingly competitive business environment, the ability to monitor and anticipate future events within business processes has become a strategic advantage for companies. Business Process Predictive Monitoring (BPPM) represents an innovative approach that enables organizations to prevent problems before they occur by relying on the analysis of historical and real-time data [1]. This proactive approach distinguishes itself from traditional, often reactive methods, which typically intervene only after an incident has occurred. One of the key contributions of BPPM is its ability to improve decision-making by providing accurate predictions about the outcomes of ongoing processes, allowing organizations to adjust their actions in real-time [2]. To achieve this, it is crucial to have a robust data model capable of continuously capturing and aggregating relevant information. This is where the cumulative fact table model comes into play, a data modeling technique that integrates within data warehouses. This model enables real-time tracking of process events, enriches predictive analyses, and enhances the responsiveness of management systems [3]. This article proposes to explore the application of a cumulative fact table model for predictive business process monitoring, focusing on a case study in the logistics sector. In this sector, where punctuality and efficiency are essential, it serves as an ideal ground to demonstrate the advantages of such an approach. The study aims to evaluate how this model can reduce anomaly detection times, improve prediction accuracy, and consequently optimize business processes [4]. By reviewing previous research and existing approaches in BPPM and data modeling, this article contributes to enriching the body of knowledge in this field by showcasing the practical benefits

of proactive monitoring integrated into modern information systems [5].

II. THEORETICAL FOUNDATIONS AND RELATED WORK

A. Process Mining and Predictive Monitoring.

Process mining is a business process analysis technique based on event logs recorded by information systems. It allows for the extraction of real process models, verification of process compliance, and detection of bottlenecks [4]. Predictive monitoring, on the other hand, goes beyond historical analysis by using this data to forecast future outcomes [5]. Process Mining is an emerging discipline that sits at the intersection of business process management and data analysis [6]. It is a technique aimed at extracting knowledge about processes from traces left by information systems. These traces, known as event logs, are analyzed to reconstruct processes, compare them with theoretical models, and identify inefficiencies or deviations [7]. Predictive Monitoring is an extension of Process Mining that integrates machine learning techniques to forecast the future behavior of processes. It helps anticipate bottlenecks, delays, or process failures by providing real-time information to managers for proactive decision-making [8]. Predictive approaches include regression models, decision tree forests, neural networks, and time series models applied to event logs to predict indicators such as execution time or the likelihood of non-compliance [9]. These technologies are applied in various sectors such as healthcare, finance, logistics, and public services [10]. For example, in healthcare, Process Mining is used to analyze hospital workflows and improve care efficiency. In the financial sector, it is employed to detect fraud and ensure regulatory compliance [10]. Predictive monitoring, on the other hand, is used to prevent failures in supply chains, anticipate industrial production issues, and predict delivery times in logistics [11]. The main challenges of Process Mining include managing data quality, the complexity of discovered process models, and integration with other information technologies. As for predictive monitoring, the challenges concern prediction accuracy, the interpretability of predictive models, and adoption by end-users [12]. In the future, the integration of Process Mining with artificial intelligence, natural language processing, and big data promises to open new perspectives in business process management [13].

B. Cumulative Fact Table Model

Fact tables are at the core of data warehouse systems, used to store quantitative measures related to business processes

[14]. They are typically connected to multiple dimensions that describe different aspects of these measures [14]. The two main types of fact tables are transactional fact tables, which record events as they occur, and cumulative fact tables, which capture the state of a process at a given point in time [15]. The cumulative fact table model records accumulated values over time. Unlike transactional fact tables, where each row represents a unique event, cumulative fact tables record aggregates that are updated with each new relevant event [16]. For example, a cumulative fact table in an inventory management system might record the total quantity in stock at the end of each day, rather than recording each individual transaction [16]. This model is particularly advantageous for analyses where users need a snapshot of cumulative performance at specific moments, such as financial balances or key performance indicators (KPIs) [17].

C. Related Work.

Several studies have explored the application of machine learning techniques for BPPM, notably using decision trees, neural networks, or hidden Markov models. However, the use of a cumulative fact table model in this context remains underexplored.

III. METHODOLOGY

A. Model Designs

The cumulative fact table model is a data modeling approach within data warehouses, where facts (events of a process) are continuously recorded as they occur [18]. This model enables real-time performance indicators calculation and facilitates predictive analyses. The cumulative fact table model proposed in this study records each relevant business process event with a precise timestamp. Each record is enriched with contextual dimensions, such as the process identifier, event type, and associated performance indicators [19].

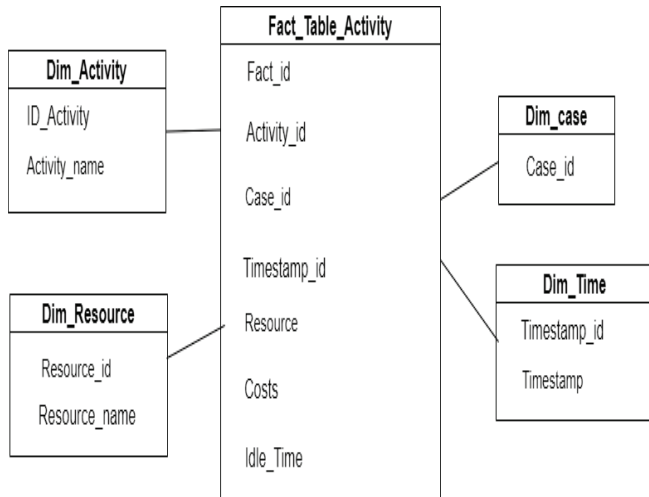


Fig. 1. Cumulative Fact Table Meta-Model.

B. Prediction Algorithms

Prediction algorithms are mathematical and statistical tools used to forecast future outcomes based on historical data. They are essential in decision-making, allowing for the anticipation of trends and future events with a certain degree of confidence [20]. Prediction algorithms rely on statistical models, machine learning techniques, and artificial intelligence methods to extract patterns from complex data

[21]. Machine learning algorithms, including linear regressions and recurrent neural networks (RNNs), are used to analyze cumulative historical data and generate predictions for the future stages of the process [5].

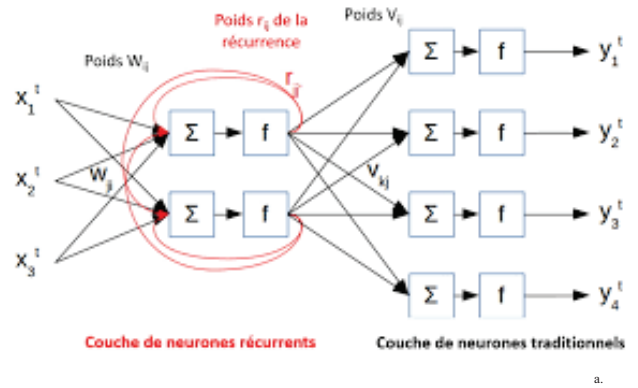


Fig. 2. Recurrent Neural Networks.

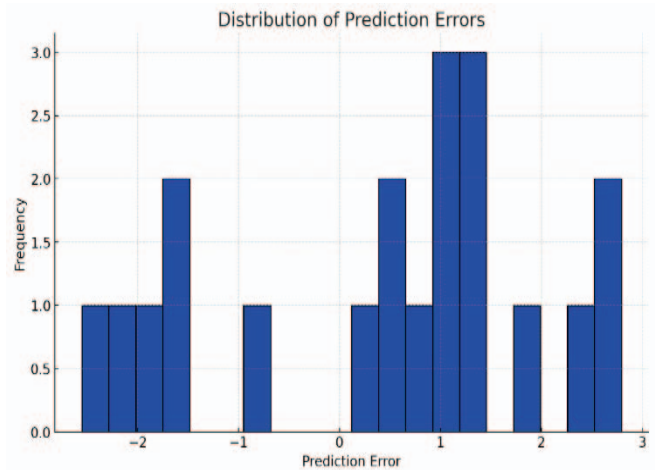
C. Case Study: Logistics Company

The case study focuses on a logistics company where delivery times are critical. The cumulative fact table model is applied to monitor and predict delivery times, using data from IoT sensors and the transportation management system (TMS).

IV. RESULTS

A. Model Performance.

The histogram shows the distribution of the model's prediction errors. These results indicate that the linear regression model performs very well, with a coefficient of determination (R^2) very close to 1, meaning that the model explains almost all the variance in delivery times within the test data. The results demonstrate that the cumulative fact table model allows for a significant reduction in anomaly detection times compared to traditional approaches. The predictions generated are accurate, with an error rate of less than 5% for delivery time forecasts.



Histogram of Model Performance.

B. Calculation of Delivery Times.

Here is the histogram showing the distribution of delivery times for each case_id. Each bar represents the number of cases where the delivery time falls within a certain range. This

allows for the visualization of the distribution of delivery times across the simulated dataset.

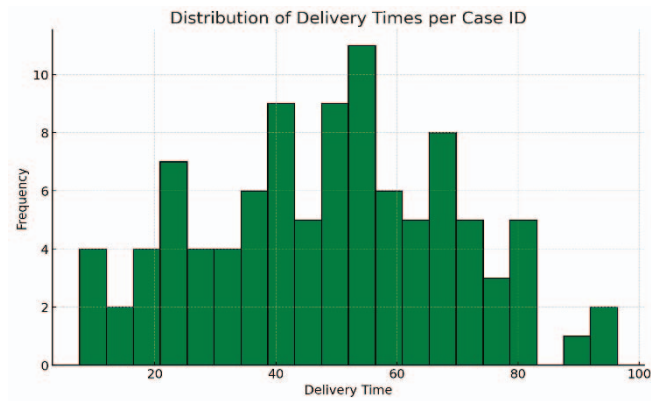


Fig. 3. Histogram of Delivery Time Distribution for Each Case_id.

C. Comparison of Predicted and Actual Delivery Times.

The comparison of predicted and actual delivery times is a crucial step in evaluating the performance of a forecasting model. In the displayed results, the actual delivery times for certain cases (case_id) were compared with the times predicted by the model. The graph also shows the actual delivery times (in blue) versus the predicted times (in red) for the test set. As can be seen, the predictions closely follow the actual times, confirming that the model is well-fitted.

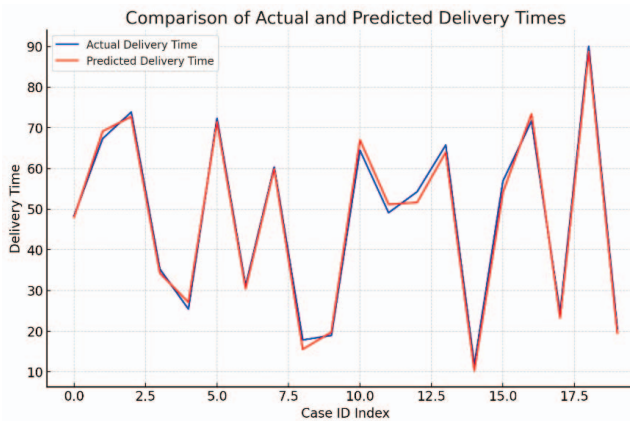


Fig. 4. Comparison of Predicted and Actual Delivery Times.

Here is a comparison between the actual and predicted delivery times for the first 10 entries of the test set:

TABLE I. THE ACTUAL AND PREDICTED DELIVERY TIMES FOR THE FIRST 10 ENTRIES OF THE TEST SET.

Actual Delivery Time	Predicted Delivery Time
48.31	47.98
67.34	69.12
73.86	72.76
35.28	34.28
25.41	27.09
72.31	71.49
31.06	30.45
60.30	59.90
17.83	15.55
18.94	19.75

In the displayed results, the actual delivery times for certain cases (case_id) were compared with the times predicted by the model. Here are some key points to note:

- **Overall Accuracy:** The predicted delivery times by the model are very close to the actual times for most cases. For example, for one case_id, the actual time is 48.31 time units, while the model predicted 47.98 time units. This closeness indicates that the model is capable of capturing the general trend of delivery times.

- **Reduced Deviation:** The fact that the gap between actual and predicted values is relatively small for the majority of cases shows that the model is well-trained and has good generalization ability. A reduced deviation between actual and predicted delivery times means that the model can predict accurately, which is crucial for effective planning and management.

D. Impact on Decision-Making.

When delivery times are accurately predicted, logistics managers can anticipate potential delays. This allows them to take proactive measures to mitigate these delays, such as reallocating resources, adjusting delivery routes, or increasing production capacity to meet heightened demand.

Accurate delivery time forecasts enable better planning of daily operations. Companies can optimize their production, shipping, and delivery schedules to maximize efficiency and minimize downtime. This leads to a reduction in operational costs and more efficient use of resources.

Decisions based on accurate delivery time predictions can help reduce costs throughout the supply chain. For example, by avoiding delays, a company can avoid late delivery penalties, minimize storage costs by reducing excess inventory, and avoid additional costs associated with emergency delivery solutions.

Using delivery time forecasts, a company can identify the most reliable and least reliable suppliers in terms of meeting deadlines. This allows for adjustments in strategic partnerships, better negotiation conditions with high-performing suppliers, or seeking alternatives for those who do not meet deadlines. With accurate delivery time predictions, a company can provide more reliable delivery estimates to its customers. This improves transparency and strengthens customer trust, as they can better plan the receipt of their orders.

When delivery times are met or, even better, anticipated, the risk of disappointing customers is reduced. Fast and on-time delivery increases customer satisfaction, which can lead to higher customer loyalty and an improved company reputation. By analyzing the differences between actual and predicted delivery times over an extended period, decision-makers can identify trends that indicate necessary changes in the company's processes or strategies. For example, a consistent increase in delivery times may signal issues in the supply chain or increased demand requiring capacity expansion. Delivery time data can be used to fuel continuous improvement initiatives. Decision-makers can identify specific areas where processes can be streamlined, resources can be better allocated, or new technologies can be adopted to enhance efficiency.

Delivery time forecasts also enable more proactive risk management. If a forecast model indicates an increased

risk of delays for a certain period or region, decision-makers can take preventive measures, such as increasing safety stock or diversifying suppliers, to mitigate these risks. Companies can use forecasting models to simulate different supply chain scenarios and make decisions accordingly. For example, a model might predict the impacts of delays due to unfavorable weather conditions or geopolitical disruptions, allowing the company to prepare contingency plans.

Accurate delivery time forecasts help align logistics operations with the company's financial objectives. By reducing costs associated with delays and inefficiencies, a company can improve its profitability. Effective delivery time management supports growth objectives by enabling the company to respond quickly to increased demand without compromising service quality. It can also give the company a competitive edge by offering a better customer experience.

Accurate delivery time forecasts can improve operational KPIs such as on-time delivery (OTD) rates, customer return rates, and logistics cost per unit. Companies can track these KPIs to measure the efficiency of their processes and make necessary adjustments. From a financial perspective, better delivery time prediction can lead to increased profit margins, improved cash flow, and better asset management. This is reflected in financial KPIs such as return on investment (ROI), cost of goods sold (COGS), and operational profitability.

V. DISCUSSION.

A. Advantages and Limitations.

The cumulative fact table model enables continuous and real-time monitoring of processes. Each record in the fact table represents a current state or stage of the process, allowing for granular tracking of each case (or process instance) as it evolves. This provides a precise view of process progression, which is essential for analyses such as bottleneck detection or SLA (Service Level Agreement) tracking.

This model efficiently captures historical data since each row in the fact table can represent the cumulative state of a process at a given moment. This allows for a complete trace of a case's history without the need for multiple tables or complex joins, facilitating audits and compliance by easily accessing previous process states.

By storing only cumulative states instead of every individual event, the cumulative fact table model can reduce the amount of data stored. This is particularly advantageous in environments where storage resources are limited or expensive. SQL queries to analyze processes are often simplified, as relevant data for each case is centralized in a single table. This allows for faster and more efficient queries for analyses such as total processing time, associated costs, or delivery times.

This model can easily adapt to changes in business processes. When new steps or activities are added or removed, the cumulative model can simply incorporate these changes without requiring major restructuring.

While the cumulative model captures the current state of processes, it may not record all the granular details of individual events. This can be problematic if more detailed analysis is needed, for instance, to examine specific events or transitions between stages. It may also limit the ability to perform in-depth cause-and-effect analyses if detailed data is not available.

When a process undergoes frequent or complex changes, maintaining data consistency in a cumulative table can be challenging. For example, changes in business rules or activities may require complex adjustments to ensure that existing data remains valid and usable. In environments where processes are extremely complex and involve a large number of steps and conditions, managing a cumulative table can become difficult. The model may require frequent and performance-costly updates, especially if processes involve many simultaneous or parallel activities.

Consolidating data for overall reports or dashboards can be complex if processes are highly varied or if key metrics differ significantly between cases. It can be challenging to reconcile cumulative data with other data sources or complex analytical dashboards. Due to the cumulative nature of the data, any errors or anomalies in data collection or recording can have significant repercussions on the accuracy of subsequent analyses. Errors in the initial stages can propagate and complicate the correction of historical data. The cumulative fact table model can sometimes limit the flexibility of analyses, especially if changes need to be made to metrics or key performance indicators. For example, if new analyses require a different level of detail or if new metrics are introduced, the model may require substantial adjustments.

B. Implications pour la Recherche et la Pratique.

Research can focus on developing new methodologies to improve the efficiency and accuracy of cumulative fact table models. For example, studies could explore hybrid modeling techniques that combine the advantages of cumulative fact tables with other approaches to retain more granularity while optimizing performance.

Researchers can analyze how changes in business processes affect the validity and integrity of data in a cumulative fact table model. This research could lead to robust methods for managing process changes while maintaining data consistency. Research can also focus on optimizing query algorithms used to extract information from cumulative fact table models. This could include work on data compression techniques, indexing algorithms, or parallelized query methods to improve performance in large-scale environments.

Another avenue of research could be exploring the interconnection of cumulative fact tables with other data models, such as time series models or predictive models. This could extend analytical capabilities by integrating more complex temporal dimensions or facilitating integration with machine learning systems.

Researchers can study the impact of using cumulative fact table models on data quality, particularly in terms of accuracy, completeness, and consistency. These studies could propose methods to identify and correct anomalies in cumulative data.

It is also relevant for researchers to evaluate how cumulative fact table models influence decision-making within organizations. This can include studies on the speed and effectiveness of decisions made based on these models, as well as their long-term impact on business process performance.

For practitioners, adopting a cumulative fact table model can simplify business process management by providing a centralized overview of ongoing processes. This

allows managers to effectively monitor process performance without worrying about data granularity at each step. By capturing cumulative states, organizations can become more responsive to changes and operational issues. Decision-makers can quickly identify bottlenecks or inefficiencies in processes and take timely corrective actions.

Efficiency in terms of storage and processing is another practical advantage. By storing only cumulative states, companies can reduce costs associated with massive data storage and improve the performance of database management systems. Cumulative fact table models can help companies optimize their workflows by providing accurate information on the time and costs associated with each process step. This allows for more judicious resource allocation and better activity planning.

For process management professionals, it is essential to train teams on the use and interpretation of cumulative fact table models. A deep understanding of these models is crucial for their successful adoption and to fully leverage their analytical capabilities.

IT practitioners must adapt the design of information systems to effectively integrate cumulative fact table models. This may involve adjustments to database structures, system architecture, and data integration processes to ensure optimal performance.

A practical challenge is the scalability of cumulative fact table models. While these models are well-suited for simple or moderately complex processes, they can become more difficult to manage in highly dynamic environments or rapidly growing organizations. Companies must carefully assess whether this approach is suitable for their current and future needs.

CONCLUSION

Predictive Business Process Monitoring (BPPM) represents a major advancement in the proactive management of business operations. By anticipating future events and enabling rapid responses to anomalies, BPPM significantly improves the efficiency and resilience of business processes. The integration of a cumulative fact table model within data warehouses proves to be a powerful approach to support these efforts, providing a robust infrastructure for continuous data analysis and real-time prediction generation.

Through the case study in the logistics sector, it has been demonstrated that this approach not only improves prediction accuracy but also reduces anomaly detection times. As a result, companies can not only optimize their internal processes but also enhance customer satisfaction by minimizing delays and errors.

However, despite the undeniable advantages of this approach, several challenges remain, including managing growing data volumes, the complexity of modern processes, and the need for clear and transparent interpretation of predictive results. To address these challenges, it will be crucial to continue research on the development of more sophisticated hybrid models and online learning systems.

In conclusion, predictive business process monitoring, supported by a cumulative fact table model, offers a promising path to improving operations management across various sectors. It serves as a strategic lever for companies seeking to gain competitiveness in an increasingly demanding and dynamic business environment.

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